

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and maintained backend AI applications using **Python** and integrated **LangChain** for orchestrating LLM interactions, processing approximately 1,500 requests per hour.
- Implemented prompt engineering techniques and experimented with multiple AI models, including fine-tuning a transformer model for improved response relevance on a 50k-record dataset.
- Designed and deployed REST APIs using **FastAPI** to serve AI model inference results, achieving a p95 latency of under **200ms** for end-user applications.
- Collaborated with data engineers to build feature pipelines for AI model training, ensuring data quality and availability for downstream ML workflows.
- Applied **MLOps** practices for model deployment to **AWS SageMaker**, including setting up monitoring for performance drift and retraining triggers.
- Built intuitive, accessible, and standards-compliant UIs for GenAI applications using **React** and **TypeScript**, integrating with backend AI services via **GraphQL**.
- Containerized AI services using **Docker** and deployed them to an **AWS EKS** cluster, managing configurations with **Helm** for repeatable deployments.

SKILLS

Languages – Python, Java, SQL, JavaScript, TypeScript, R, HTML, CSS

AI/ML Frameworks & Libraries – TensorFlow, PyTorch, scikit-learn, LangChain, Semantic Kernel, pandas, NumPy

Cloud Platforms & Services – AWS (SageMaker, EC2, S3, Lambda, EKS, CloudWatch), Azure ML

Backend & API Technologies – Spring Boot, FastAPI, Flask, REST, GraphQL, Node.js

Frontend Frameworks – React, Angular, Vue.js

DevOps & Orchestration – Docker, Kubernetes, Helm, Jenkins, Git, CI/CD

Databases – PostgreSQL, MySQL, Oracle

PROJECTS

AI-Powered Customer Support Assistant (MSc Project)

- Developed a RAG-based chatbot using **LangChain** and an **OpenAI** LLM to answer technical queries, integrating with a vector database (Pinecone) for knowledge retrieval.
- Engineered Python scripts to preprocess and embed documentation into vector embeddings, enabling the AI to provide contextually relevant answers, handling approximately 500 unique query types.
- Built a simple web interface using **Flask** and **HTML/CSS** for users to interact with the AI assistant, allowing for iterative prompt refinement and testing.

Real-Time Transaction Anomaly Detection Service

- Created a Python service using **FastAPI** to expose a real-time anomaly detection model trained on financial transaction data, processing ~1,200 transactions per minute.
- Trained a **scikit-learn** Isolation Forest model on a dataset of 1.5M transactions, achieving an AUC of **0.88** and identifying 95% of simulated fraudulent activities in testing.
- Containerized the service with **Docker** and deployed it to an **AWS EC2** instance, ensuring high availability for the inference endpoint.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)